

TrES-3b

R-0375

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170609 · created 2026-07-18T18:13:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457913.7645 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.048
Transit depth (Rp/Rs)^2	0.94 +/- 0.93 [%]
Orbital Inclination (inc)	85.8 +/- 2.81
Airmass coefficient 1 (a1)	1.255 +/- 0.032
Airmass coefficient 2 (a2)	-0.22 +/- 0.044
Scatter in the residuals of the lightcurve fit is	2.54 %
Best Comparison Star	None
Optimal Aperture	3.04
Optimal Annulus	15.6
Transit Duration (day)	0.068 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.097 ± 0.048 vs 0.16309 (deviation 1.38σ)
Duration fit vs published	0.068 d vs 0.0567 d (deviation 0.76σ)
Mid-time O–C	-4.1 min
Depth SNR	2.0
Residual scatter	2.54 %

Light curve

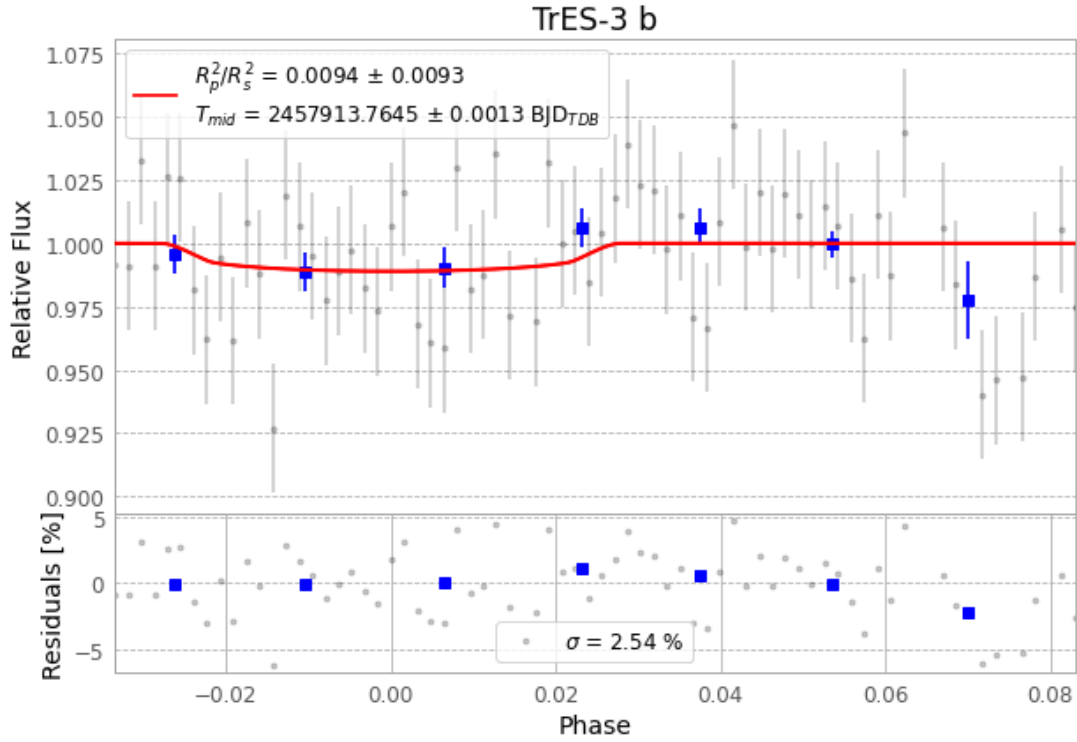


Figure 1 — FinalLightCurve_TrES-3 b_2017-06-08.png (SHA-256 bb7cca02dde7cb5e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 accbcee111d187e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3007197

Data provenance

69 FITS frames (45 MB), TRES-3170609051212.FITS → TRES-3170609085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170609.log (30ce8cbb856a...), FinalLightCurve_TrES-3 b_2017-06-08.png (bb7cca02dde7...), FinalLightCurve_TrES-3 b_2017-06-08.csv (b654df6d4e01...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 18:13Z.